



Salary Prediction Using Linear Regression

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Agenda

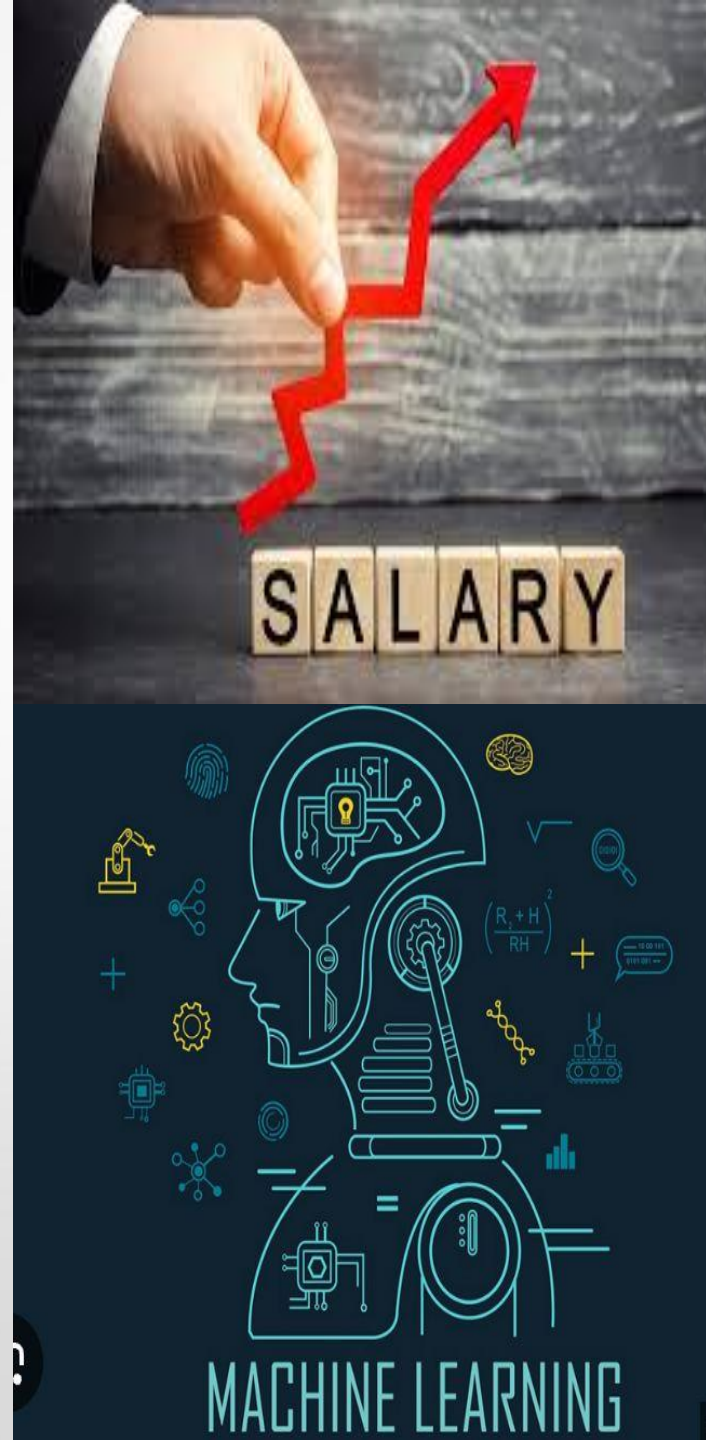
- 1. Introduction
- 2. Tools and Technologies
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- 7. Future Scope
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Flask

Introduction

- This project aims to predict the salary of an individual based on various attributes such as years of experience, education level, job role, skillset, and other relevant professional factors.
- The solution is built using machine learning techniques that analyze historical salary data and identify patterns. Once trained, the model can accurately estimate the expected salary for a new user's profile.
- This system helps organizations, HR departments, and job seekers in making informed decisions related to compensation and recruitment.

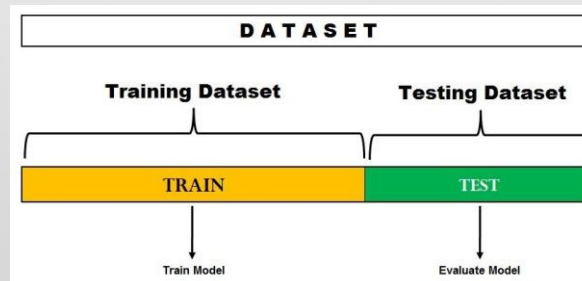
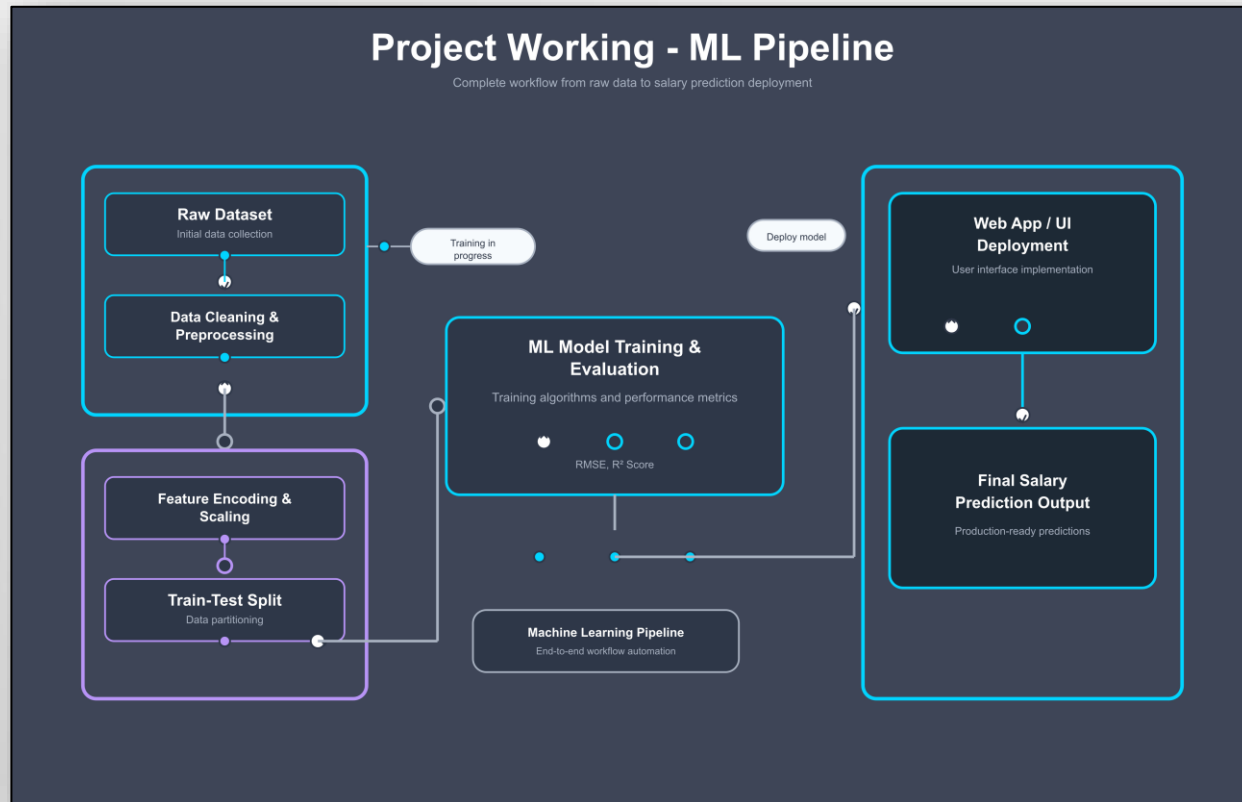


Tools and Technologies

- Python
- Scikit-Learn
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Flask Framework
- VS Code
- GitHub



Project Working



Output Screenshot 1

Salary Predictor

127.0.0.1:5000

Import favorites For quick access, place your favorites here on the favorites bar. [Manage favorites now](#)

Predict Salary

Enter details to get Salary estimate

Age
32

Gender
Female

Education Level
Master's

Job Title
Data Analyst

Years of Experience
7

Predict Salary →

👛 Predicted Salary: \$81,211.96

***A walkthrough video is shared at the project links page.**

Output Screenshot 2

Salary Predictor

127.0.0.1:5000

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Predict Salary

Enter details to get Salary estimate

Age
52

Gender
Male

Education Level
PhD

Job Title
Director

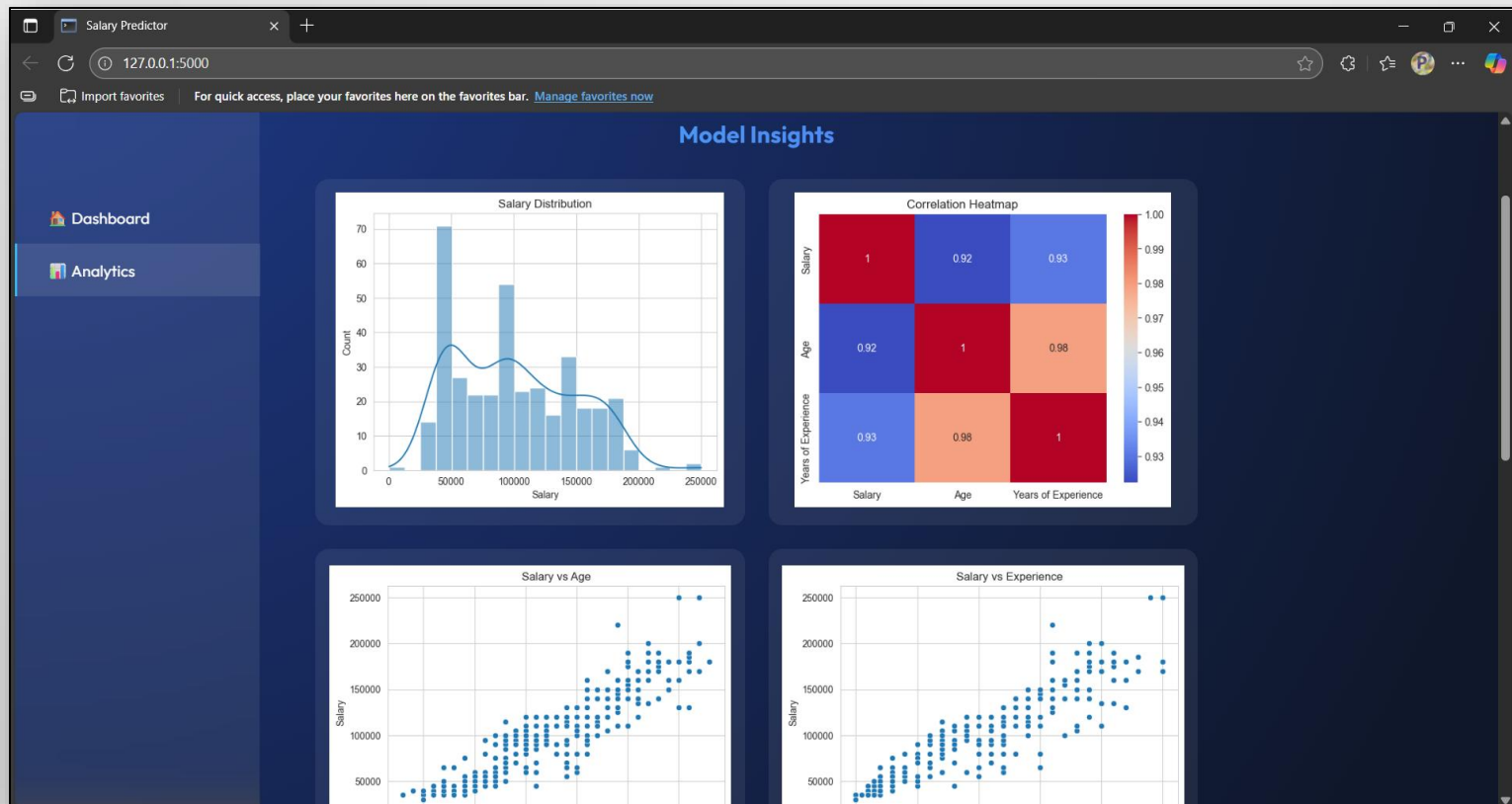
Years of Experience
25

Predict Salary →

👤 Predicted Salary: \$2,01,134.04

***A walkthrough video is shared at the project links page.**

Output Screenshot 3



***A walkthrough video is shared at the project links page.**

Applications

- **Human Resource Departments** to estimate competitive salaries.
- **Job Seekers** to evaluate expected salary for job offers.
- **Recruitment Agencies** for assessing market compensation trends.
- **Educational and Career Guidance Platforms** to help students set professional goals.
- **Companies** for compensation benchmarking and workforce planning.

Problems Faced & Solutions

Challenge	Explanation	Solution
Missing or inconsistent data	Real-world datasets contain incomplete values and noise	Applied data cleaning, imputation, and normalization
Categorical feature handling	Model cannot directly process non-numeric features	Used One-Hot Encoding for categorical variables
Model overfitting	Model performed better on training data but poorly on testing data	Hyperparameter tuning, cross-validation applied
Deployment issues	Model required proper environment setup	Created requirements.txt and tested in virtual environment

Future Scope

- Integrate **more advanced ML models** such as XGBoost or Neural Networks to improve accuracy.
- Develop a **mobile application** for wider accessibility.
- Enable **real-time salary prediction** with live job market datasets.
- Incorporate **additional features** like company size, location-based cost index, and skill proficiency rating.
- Launch as an **enterprise HR integration tool** for automated recruitment workflows.

Project Link

- GitHub Repository:
https://github.com/PRo8o1/Salary_Prediction
- Walkthrough Video :
https://drive.google.com/file/d/1sPmP5ZuukZLnRc7lMEXAr_YdpcdyXPVR/view?usp=sharing